

Assignment 4: Becoming an Independent Data Scientist

1. Region and Domain

State the region and the domain category that your data sets are about.

The **NYC Yellow Taxi Dataset** is a public dataset that provides detailed records of taxi trips in New York City. Here is a breakdown of the region and domain category:

Region:

The dataset covers taxi trips specifically within **New York City (NYC)**, including all five boroughs—Manhattan, Brooklyn, Queens, The Bronx, and Staten Island. Some trips may extend slightly outside of NYC for destinations such as nearby airports or surrounding areas.

Domain Category: The domain of the dataset falls under transportation, specifically in the area of urban mobility and transportation analytics. It is often used for studying traffic patterns, transportation demand, ride-hailing services, public policy, and other areas of interest like urban planning, economics, and sustainability.

The data includes information such as pickup and dropoff locations, times, fare amounts, and passenger counts, among other features.

2. Research Question

You must state a question about the domain category and region that you identified as being interesting.

How have the **number of passengers (passenger density)** in the **yellow taxis** for **NYC** for **different TLC Taxi Zones** (e.g., JFK Airport, Midtown Center, Penn Station/Madison Sq West etc.) in the month **April** of the year **2023** differ over with respect to different

1. **zones**
2. **times (hours)** in a day
3. **pickup** vs. **dropoff**
4. **weekdays** vs. **weekends?**

3. Links

You must provide at least two links to publicly accessible datasets. These could be links to files such as CSV or Excel files, or links to websites which might have data in tabular form, such as Wikipedia pages.

Here are couple of publicly accessible sources where we can find the **NYC Yellow Taxi Dataset** (for the year **2023** and month **April**):

1. NYC OpenData

https://data.cityofnewyork.us/Transportation/2023-Yellow-Taxi-Trip-Data/4b4i-vvec/data_preview

2. TLC Trip Record Data

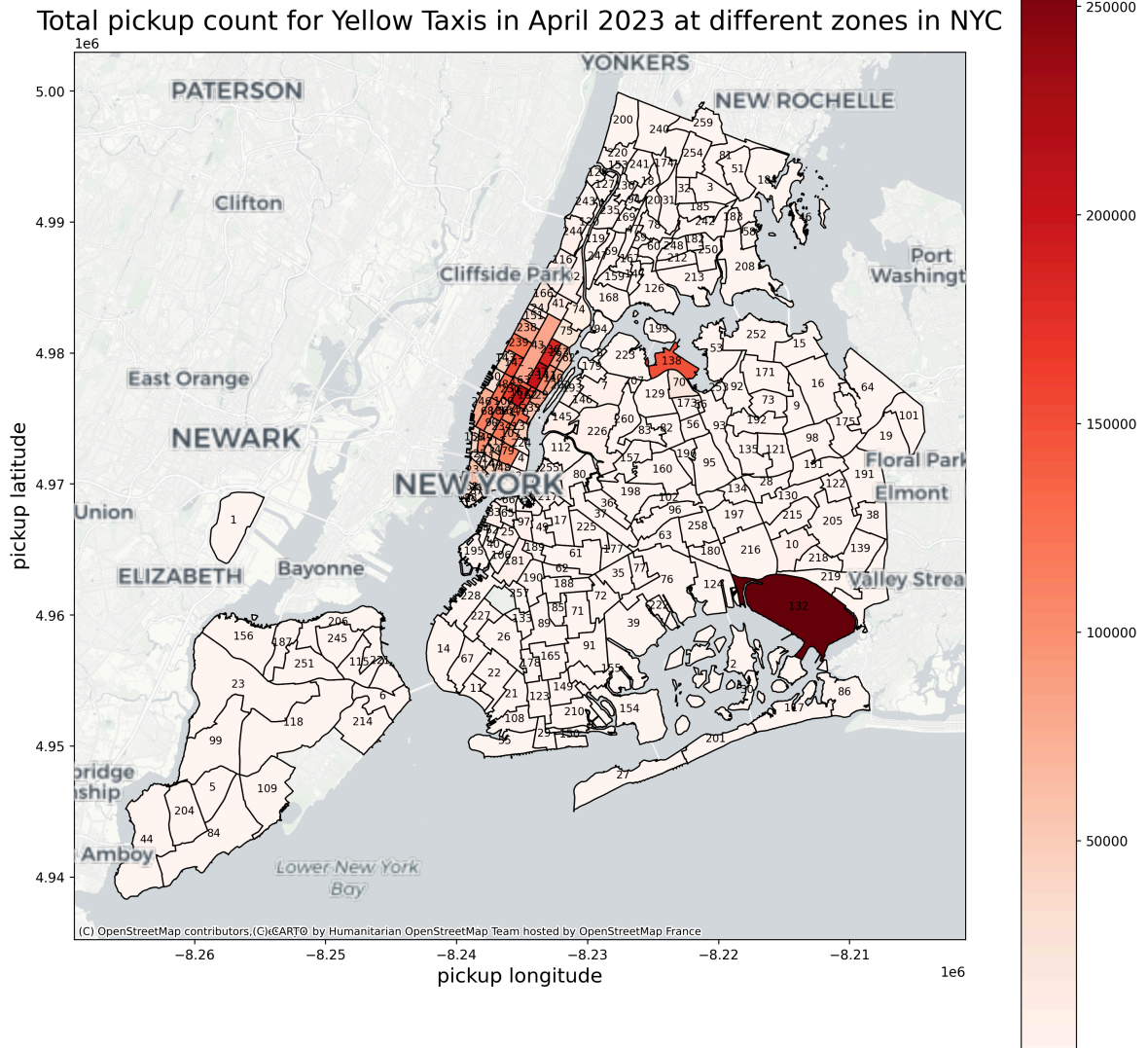
- <https://www.nyc.gov/site/tlc/about/data-and-research.page>
- https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_yellow.pdf

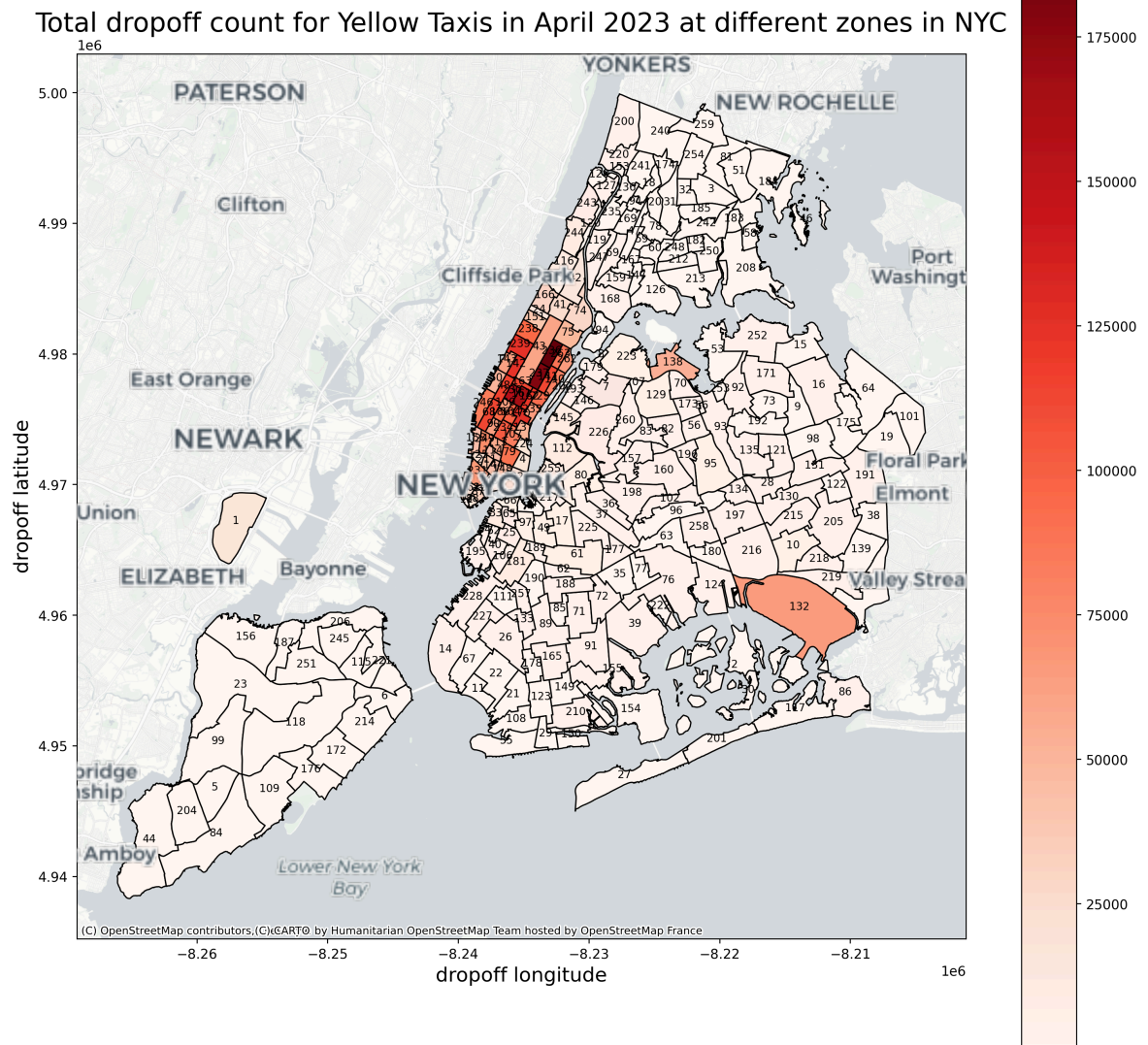
4. Image

You must upload an image which addresses the research question you stated. In addition to addressing the question, this visual should follow Cairo's principles of truthfulness, functionality, beauty, and insightfulness.

Total Pickup vs. Dropoff Count at different Zones

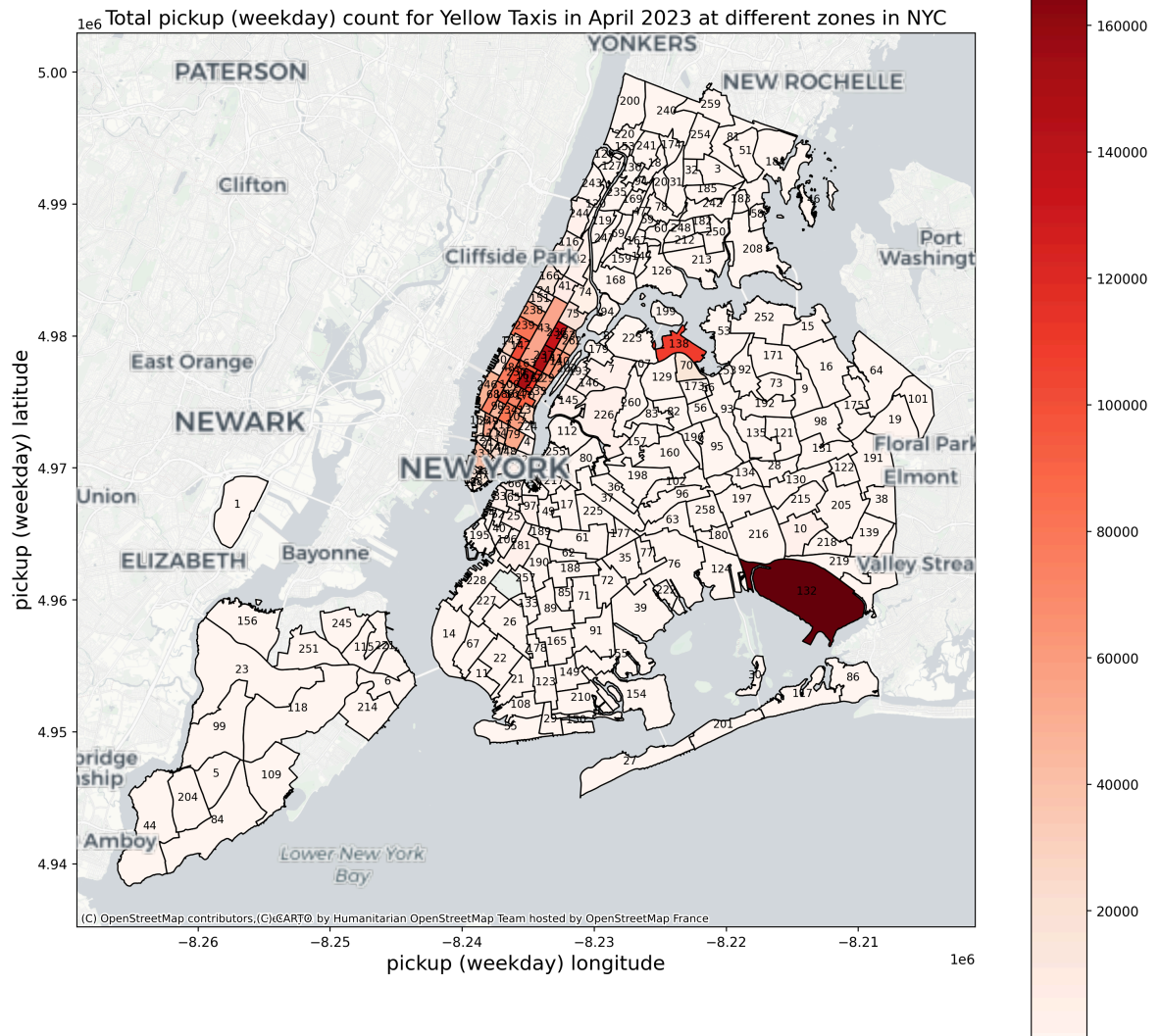
In [5]: `## Total Pickup vs. Dropoff Count at different Zones`

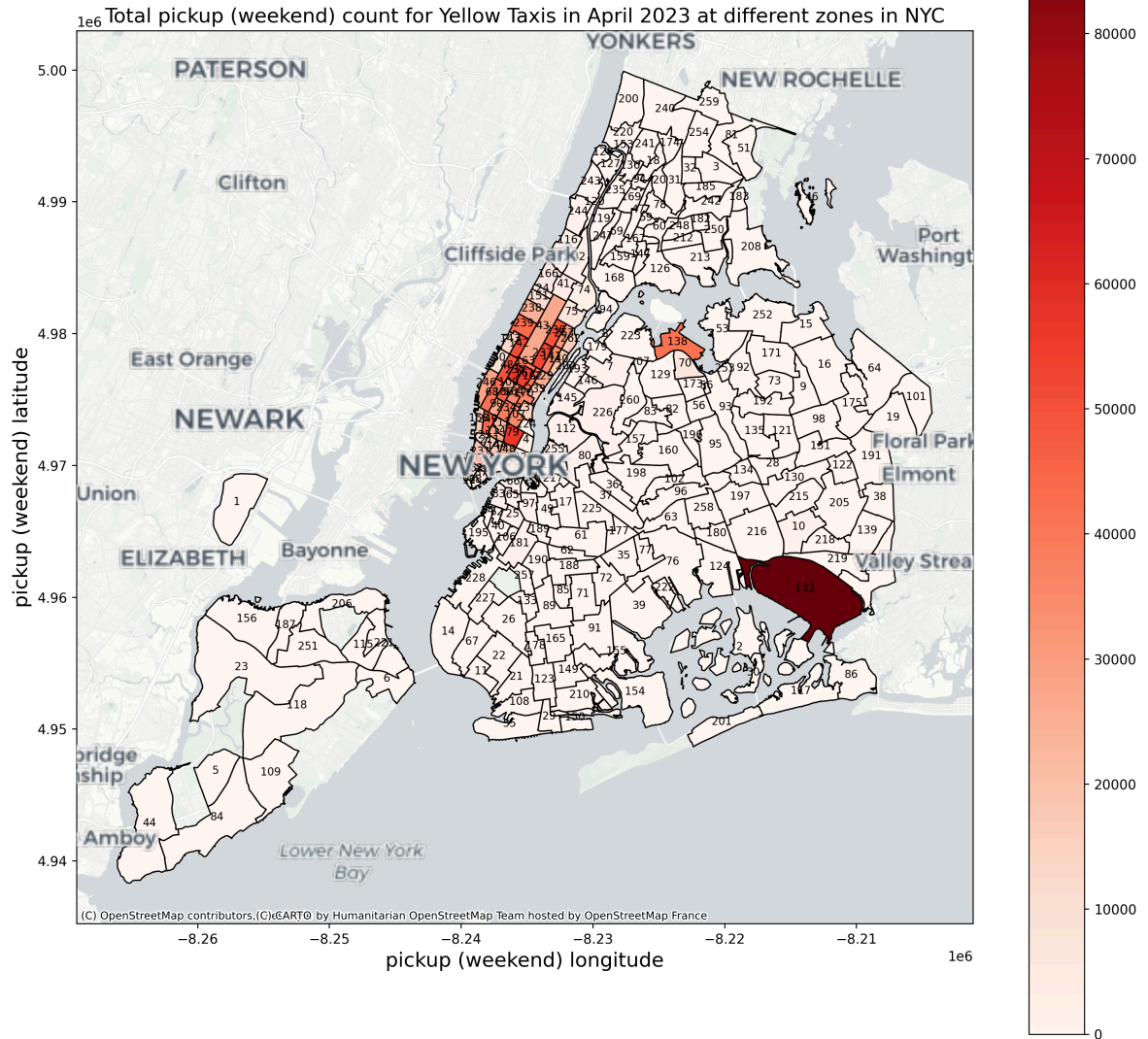




Total Pickup Count at different Zones in the Weekdays vs. Weekends

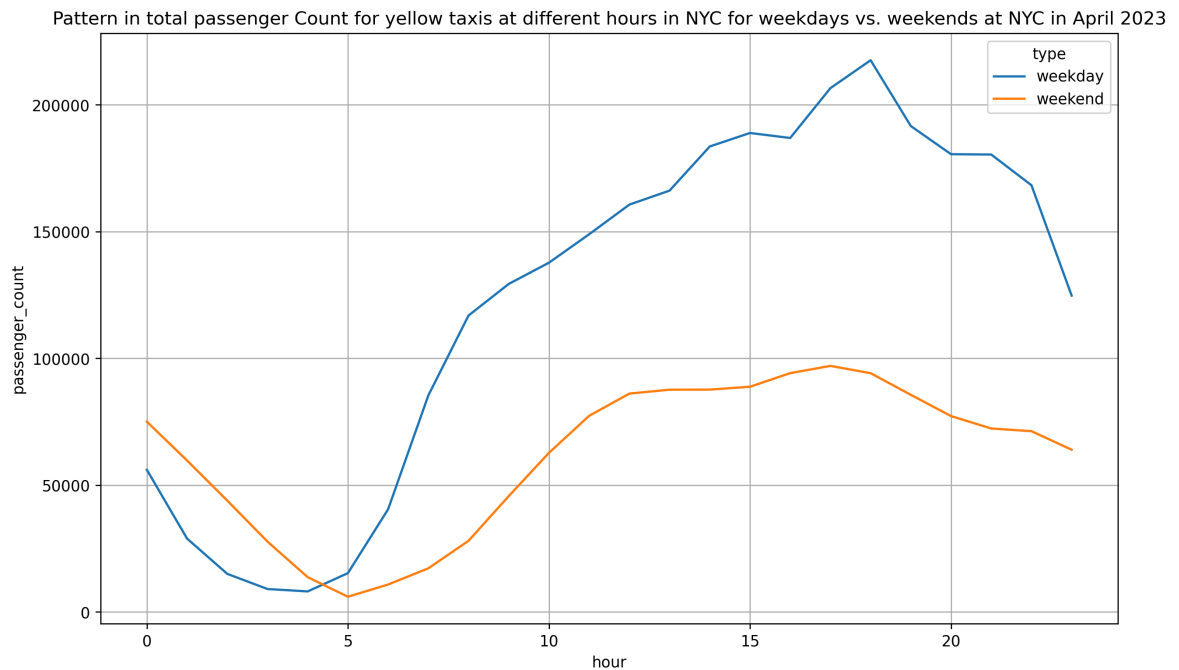
```
In [35]: ### Total Pickup Count at different Zones in the Weekdays vs. Weekends
```





Total Pickup Passenger Count at different hours of a day in the Weekdays vs. Weekends

In [28]: `### Total Pickup Passenger Count at different hours of a day in the Weekdays vs.`



5. Discussion

You must contribute a short (1-2 paragraph) written justification of how your visualization addresses your stated research question.

Total Pickup vs. Dropoff Count at different Zones

As can be seen from the geospatial visualizations, the zones with maximum total passenger count (marked with darker color) for pickup (we get similar zones for dropoff too) are

- JFK Airport (LocationID 132)
- Midtown Center (LocationID 161)
- Upper East Side South (LocationID 237)

In [33]: `### Total Passenger Pickup Count at different Zones`

Out[33]:

	OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	borough		
	126	132	0.245479	0.002038	JFK Airport	132	Queens	MULTIPOINT((101 1
	155	161	0.035804	0.000072	Midtown Center	161	Manhattan	MULTIPOINT((9909
	229	237	0.042213	0.000096	Upper East Side South	237	Manhattan	MULTIPOINT((9935
	228	236	0.044252	0.000103	Upper East Side North	236	Manhattan	MULTIPOINT((9
	222	230	0.031028	0.000056	Times Sq/Theatre District	230	Manhattan	MULTIPOINT((9886
	136	142	0.038176	0.000076	Lincoln Square East	142	Manhattan	MULTIPOINT((9893
	132	138	0.107467	0.000537	LaGuardia Airport	138	Queens	MULTIPOINT((102
	156	162	0.035270	0.000048	Midtown East	162	Manhattan	MULTIPOINT((9920
	178	186	0.024696	0.000037	Penn Station/Madison Sq West	186	Manhattan	MULTIPOINT((9866
	157	163	0.034177	0.000041	Midtown North	163	Manhattan	MULTIPOINT((9900

Total Pickup Count at different Zones in the Weekdays vs. Weekends

As can be seen from the second pair of geospaial images, the density of the passengers for pickup follows similar patterns at different zones for weekdays and weekends, with

the main difference being that the total number of passengers picked up at different zones in the weekdays are **much higher in the weekdays than in the weekends**, as expected.

Total Pickup Passenger Count at different hours of a day in the Weekdays vs. Weekends

As can be seen from the line plots, the peak hour for the total pickup count of the passengers is in between 5 — 7 **PM** in the evening, whereas the total count drops significantly in the weekends, the trend being more or less the similar.

In []:

